

Project name: The hospital Booking System

Team name: Aura

members (id, name):

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PROBLEM

Patients currently rely too much on hospital staff, members, calls, and even in-person visits to check the availability of the hospital rooms and book appropriate doctors. A long queue, unpredictable waiting hours, and unavailability of doctors after a long wait can result in the disappointment of patients (customers). It is becoming more serious these days as hospitals are overcrowded than they used to. To solve these major problems, we design a system where patients can check the availability of rooms and make appointments with doctors effortlessly via the system online.

GATE 1 — REACHABLE USERS

1. นพ.คนพัฒน์ พรพิพิธ - Family Medicine Physician, Phramongkutklao Hospital – 24 Aug 2026
2. นางสาวอัมรา ตาแก้ว - Housewife – 24 Aug 2026
3. นางพิศ ปัญญาวรรณ – Housewife – 24 Aug 2026
4. นางฐานมาศ กิติภักย์พิบูลย์ – Housewife – 24 Aug 2026
5. นางสาว พิมพ์พิชชา คุณมาศ – Collegian – 24 Aug 2026

GATE 2 — MVP SHAPE

3–5 planned functional requirements:

FR-1: User Register and Login

The system should allow all the patients to create their personal accounts and sign in before using the system for the appointment, and queue information must be linked to the individual patient.

FR-2: View Doctor Information and Availability

The system should allow all the patients to look for doctors, their specializations, and available appointment dates and times before booking. This helps patients find a relevant doctor without checking availability manually.

FR-3: Book an Appointment

The system should allow patients to pick up a doctor, date and time, and confirm an appointment online. The system should prevent a patient from booking a time slot that's already taken and should create a token number after the booking.

FR-4: Check Current Token Queue

The system should allow all patients to view their personal number, the current queue, and the status of their appointment. This helps patients understand their position, in the queue, and monitor each of their progress.

FR-5: Manage Token Numbers and Patient Queue

The system should allow the admins to manage the patients' queue to call the patient by their individual token numbers and update token status through the outpatient process. Token status includes Waiting, Call, Operation, Completed and Cancelled.

Which FR carries a real rule or decision (not plain CRUD)?

FR-5: Manage Token Numbers and Patient Queue carries the business rules of the MVP. The system must determine the queue to assign token numbers to confirmed appointments to allow the admin to call the next patient and update the token status as the patient moves through the process.

FR-3: A time slot that is already taken cannot be booked by another patient.

What data has to persist between sessions?

The system needs to persist the following data:

- Patient account information (personal)
- Doctor information and their specializations
- Doctor schedules and availability of appointment slots
- Appointment information
- Token numbers
- Token status
- Patient queue information
- Appointment history

This persistent data is required so that patients and admins can continue to access and manage appointments and queue information after logging out or starting a new session.

GATE 3 — DATA ACCESS

Our Hospital OPD Booking System requires detailed information about the doctors, their schedules, appointments, and token numbers.

For the MVP, we do not ask patients to access their unnecessary medical records. We will be using publicly available information from real hospitals and use it as a reference to create samples for doctors, patients, schedules, appointments, and token numbers.

We learn how real hospitals OPD booking System works. Depending on what we learn, we are going to design our own booking processes and use sample data to develop the working system.

PROCESS & TEAM

Incremental/Iterative model (or Agile) — Because a hospital booking system has many requirements that change over time and needs feedback and review from people including doctors, admin staff and patients. It helps lower risk by delivering working parts of the system.